Circulation Journal
Circ J 2026; **90:** 305–314
doi:10.1253/circj.CJ-25-0097

ORIGINAL ARTICLE

Surgery

Surgical Results for Anomalous Aortic Origin of Coronary Artery

— Is Right Side Prompt Surgery Necessary? —

Sangil Yun, MD; Joowon Lee, MD; Jae Gun Kwak, MD, PhD;
Sang Yun Lee, MD, PhD; Woong-Han Kim, MD, PhD

Background: This study evaluated early and mid-term clinical outcomes of surgical correction for anomalous aortic origin of a coronary artery (AAOCA) and expansion of surgical indications beyond current guidelines, particularly for asymptomatic patients with anomalous aortic origin of the right coronary artery (AAORCA).

Methods and Results: Between December 2004 and July 2023, 34 patients underwent surgery for AAOCA. Surgical indications included evidence of myocardial ischemia and high-risk anatomy. AAOCA was assessed pre- and postoperatively using imaging and functional studies. Early and mid-term outcomes were evaluated retrospectively. AAORCA was the predominant type (n=32; 94.1%), and 32 of 34 patients (94.1%) underwent unroofing. Five (14.7%) asymptomatic AAORCA patients had a history of Kawasaki disease. There were no surgical mortalities or coronary artery-related complications. Of 12 patients with symptoms or signs likely related to the coronary artery in the early postoperative period, 2 had persistent symptoms until the last follow-up. During follow-up, 2 patients had suspicious mild coronary stenosis on computed tomography, and 1 had decreased stress perfusion on a myocardial perfusion scan. Among patients with preoperative abnormalities, 92.3% exhibited postoperative functional improvement.

Conclusions: Surgical treatment of AAOCA, primarily through unroofing, is safe and effective, with favorable early and mid-term outcomes. Our findings support consideration of surgery for asymptomatic AAORCA patients with high-risk anatomy due to the potential risk of sudden cardiac events and the substantial benefits of the procedure.

Key Words: Anomalous aortic origin of a coronary artery; Anomalous aortic origin of the right coronary artery; Coronary vessel anomalies; Mucocutaneous lymph node syndrome; Unroofing

Coronary artery anomalies, occurring in 0.2–1.2% of the population, based on previous reports from the North American, can lead to myocardial ischemia and sudden death, particularly among young athletes.[1,2] Despite its rarity, anomalous aortic origin of a coronary artery (AAOCA) is one of the most clinically significant types of coronary artery anomalies. Many patients may remain asymptomatic, but there is a risk of sudden death if the artery takes an intramural course between the aorta and pulmonary artery.[3] Current guidelines recommend surgery for symptomatic patients with AAOCA and asymptomatic patients with test results indicating ischemia, but not for asymptomatic patients without such results.[4,5] The aim of this study was to analyze the surgical outcomes for AAOCA at Seoul National University Children's Hospital and explore the potential for expanding the indications for surgery.

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Methods

Ethics Approval

This study was conducted in accordance with the Declaration of Helsinki and was approved by the Institutional

Received February 10, 2025; revised manuscript received June 30, 2025; accepted July 4, 2025; J-STAGE Advance Publication released online September 12, 2025 Time for primary review: 37 days

Department of Thoracic and Cardiovascular Surgery, Seoul National University Children's Hospital, Seoul National University College of Medicine, Seoul (S.Y., J.G.K., W.-H.K.); Department of Pediatrics, Seoul National University Bundang Hospital, Seoul National University College of Medicine, Seongnam-si (J.L.); and Department of Pediatrics, Seoul National University Children's Hospital, Seoul National University College of Medicine, Seoul (S.Y.L.), Republic of Korea

Mailing address: Jae Gun Kwak, MD, PhD, Department of Thoracic and Cardiovascular Surgery, Seoul National University Children's Hospital, Seoul National University College, 101 Daehak-ro, Jongno-gu, Seoul 03080, Republic of Korea. email: switch.surgeon@gmail.com

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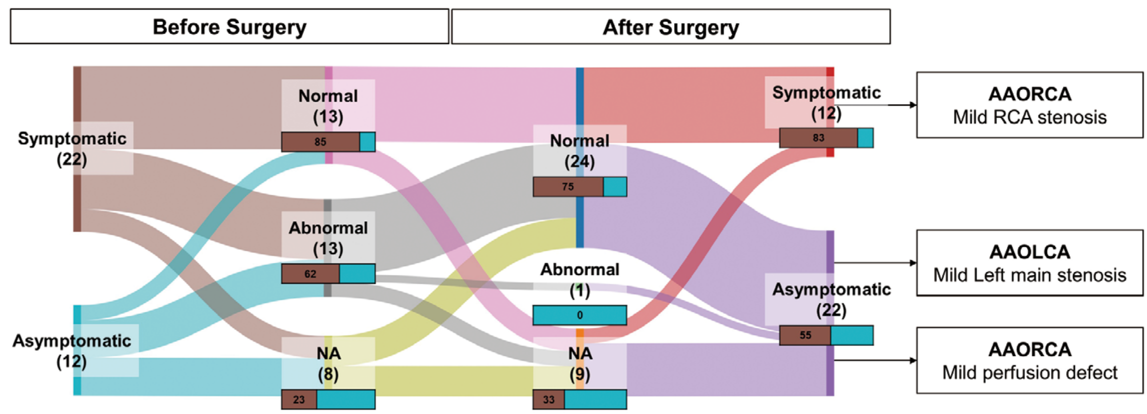
ISSN-1346-9843



Central Figure

Review Board of Seoul National University Children's Hospital (Approval no. H-2308-094-1458; date of approval: August 21, 2023). Due to the retrospective design of the study, the requirement for individual consent was waived.

Study Population

Between December 2004 and July 2023, 34 patients underwent surgery for AAOCA at Seoul National University Children's Hospital; all procedures were performed by any of the two surgeons (K.W.H. or K.J.G.). The diagnosis was initially assessed by preoperative echocardiography and computed tomography (CT), classified per the Congenital Heart Surgery Nomenclature and Database Project.[6] Clinical evaluation involved several imaging studies, such as cardiac magnetic resonance imaging (CMR) and coronary angiography, alongside functional assessments, such as the treadmill test (TMT) or myocardial single photon emission CT (SPECT). These studies were performed selectively based on preoperative urgency and patient age.

A surgical intervention for AAOCA was indicated not only for those with stress-induced myocardial ischemia but also for asymptomatic patients with high-risk anatomical features, including an intramural course and orifice anomalies (slit-like or acute-angle takeoff).

Surgical Techniques and Strategy

Coronary artery unroofing was performed in patients with an intramural course of the coronary artery, whereas those with acute-angle takeoff anomalies underwent coronary artery reimplantation. All procedures were conducted via median sternotomy with moderate hypothermic cardiopulmonary bypass and antegrade cardioplegic arrest using del Nido cardioplegia solution, with retrograde cardioplegic perfusion through the coronary sinus performed as necessary. After cross-clamping, the aorta was transected, and the orifices and courses of the coronary arteries were meticulously inspected. The techniques for coronary artery unroofing have been detailed previously.[7,8] The intramural course was unroofed, and the resected margin was secured using eversion sutures, with multiple simple interrupted stitches or continuous sutures using 7–0 or 8–0 polypropylene for re-endothelialization (**Figure 1**). Aortic valve resuspension was performed at the surgeons' discretion to prevent aortic regurgitation (AR).[9] For reimplantation, the ostium of the anomalous coronary artery was excised along with the adjacent aortic tissue and reimplanted in an anatomically adequate area of the aortic sinus. The coronary artery excision site was reconstructed using the patient's own aortic tissue.

Evaluation of Early and Mid-Term Clinical Outcomes

The early clinical outcomes assessed were hospital stay, intensive care unit (ICU) stay, operative mortality, and complications. The mid-term clinical outcomes evaluated were the need for reintervention for the coronary artery and abnormal findings from imaging or functional studies.

Operative mortality was defined as death occurring within 30 days after surgery. Coronary artery-related complications included myocardial infarction, sustained ventricular tachycardia, and ventricular fibrillation. All patients underwent continuous electrocardiogram monitoring until discharge. Myocardial infarction events were confirmed through medical records, evidenced by elevated biomarkers and electrocardiographic alterations according to the universal definition of myocardial infarction.[10]

Pre- and Postoperative Investigations

Echocardiography and electrocardiography were routinely conducted preoperatively. Echocardiography was repeated in 97.1% (33/34) of patients a median of 4 days (interquartile range [IQR] 3–5 days) postoperatively to assess cardiac function, coronary artery flow, coronary ostia, AR, and other treated anomalies. During follow-up, echocardiography was performed at 6 months (median 0.6 years; IQR 0.3–1.0 years), 1 year (median 1.5 years; IQR 1.0–2.9 years), and 3 years (median 2.7 years; IQR 1.7–5.0 years), with additional echocardiography as needed. Pre- and postoperative imaging and functional studies, including cardiac CT, CMR, coronary angiography, TMT, and myocardial SPECT, were performed, if necessary.

Statistical Analysis

Statistical analyses were performed using SPSS version 27.0 (IBM Corp., Armonk, NY, USA), RStudio version 2024.12.0+467 (RStudio, PBC, Boston, MA, USA) with R version 4.4.2 (R Foundation for Statistical Computing,

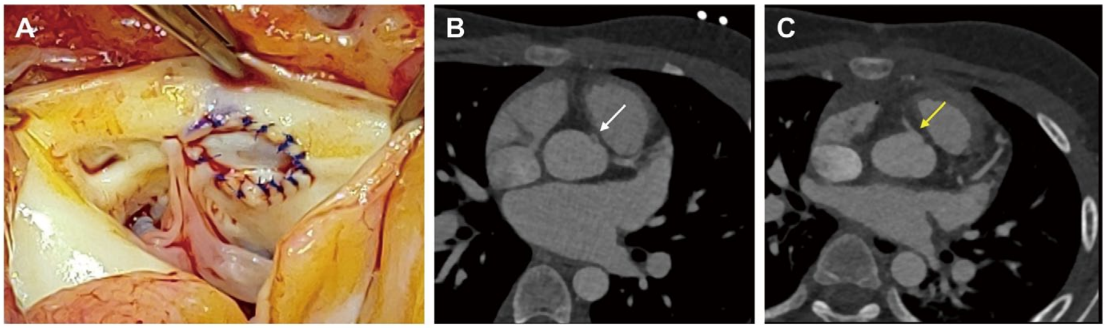


Figure 1. Coronary unroofing in an 11-year-old boy with an anomalous aortic origin of the right coronary artery, a juxtacommisural ostium, and an intramural course. (**A**) The intramural course was partially unroofed due to the juxtacommisural ostium, with the neo-ostium margin reinforced with interrupted sutures, and an aortic valve resuspension suture placed. (**B**) The diameter of the anomalous right coronary artery ostium (white arrow) was 1.3 mm (z-score –2.16), as measured by preoperative cardiac computed tomography. (**C**) The diameter of the right coronary artery neo-ostium (yellow arrow) was 3.6 mm (z-score 1.47), as measured by postoperative cardiac computed tomography.

Table 1. Baseline Characteristics

	Total (n=34)	AAOCA repair		P value
		Isolated (n=17)	Concomitant (n=17)	
Male sex	25 (73.5)	13 (76.5)	12 (70.6)	>0.999
Age at diagnosis (years)	12.0 [5.3–15.4]	12.6 [7.1–15.3]	6.6 [4.7–15.3]	0.654
Age at surgery (years)	13.4 [7.4–16.4]	13.5 [11.4–15.6]	12.9 [5.8–19.3]	0.630
Time from diagnosis to surgery (months)	5.9 [1.5–32.7]	5.9 [1.5–30.7]	5.9 [1.6–22.8]	0.823
Time <3 months	13 (38.2)	7 (41.2)	6 (35.3)	>0.999
AAOCA type				0.485
AAOLCA	2 (5.9)	2 (11.8)	0 (0.0)	
AAORCA	32 (94.1)	15 (88.2)	17 (100.0)	
LVEF <55%	3 (8.8)	1 (5.9)	2 (11.8)	>0.999
Accompanying congenital heart diseases	12 (35.3)	0 (0.0)	12 (70.6)	<0.001
Atrial septal defect	7 (20.6)	0 (0.0)	7 (41.2)	
Ventricular septal defect	3 (8.8)	0 (0.0)	3 (17.7)	
Congenital aortic stenoinsufficiency	1 (2.9)	0 (0.0)	1 (5.9)	
PAPVR	1 (2.9)	0 (0.0)	1 (5.9)	
Prior cardiac surgery	2 (5.9)	2 (11.8)	0 (0.0)	0.485
History of Kawasaki disease	5 (14.7)	2 (11.8)	3 (17.7)	>0.999
Elite athlete	3 (8.8)	2 (11.8)	1 (5.9)	>0.999
Symptomatic patients	22 (64.7)	14 (82.4)	8 (47.1)	0.071
Chest pain	16 (47.1)	11 (64.7)	5 (29.4)	
Chest discomfort	4 (11.8)	2 (11.8)	2 (11.8)	
Palpitation	3 (8.8)	1 (5.9)	2 (11.8)	
Dyspnea on exercise	1 (2.9)	1 (5.9)	0 (0.0)	
Syncope	1 (2.9)	0 (0.0)	1 (5.9)	
Cardiac CT	33 (97.1)	16 (94.1)	17 (100.0)	>0.999
CMR	1 (2.9)	0 (0.0)	1 (5.9)	>0.999
CAG	4 (11.8)	3 (17.7)	1 (5.9)	0.601
TMT	20 (58.8)	10 (58.8)	10 (58.8)	>0.999
Abnormal findings	6 (17.6)	3 (17.7)	3 (17.7)	>0.999
Myocardial SPECT	25 (73.5)	13 (76.5)	12 (70.6)	>0.999
Abnormal findings	7 (20.6)	6 (35.3)	1 (5.9)	0.085

Continuous variables are presented as the median [interquartile range]; categorical variables are presented as n (%). AAOCA, anomalous aortic origins of a coronary artery; AAOLCA, anomalous aortic origin of left coronary artery; AAORCA, anomalous aortic origin of right coronary artery; CAG, coronary angiography; CMR, cardiac magnetic resonance imaging; CT, computed tomography; LVEF, left ventricular ejection fraction; PAPVR, partial anomalous pulmonary venous return; SPECT, single-photon emission computed tomography; TMT, treadmill test.

Table 2. Operative Data

	Total (n=34)	AAOCA repair Isolated (n=17)	AAOCA repair Concomitant (n=17)	P value
Operation				0.485
Unroofing	32 (94.1)	15 (88.2)	17 (100.0)	
Reimplantation	2 (5.9)	2 (11.8)	0 (0.0)	
Approach				0.601
Full median sternotomy	30 (88.2)	14 (82.4)	16 (94.1)	
Upper partial sternotomy	4 (11.8)	3 (17.7)	1 (5.9)	
Resuspension suture	9 (26.5)	6 (35.3)	3 (17.7)	0.438
Patients with concomitant procedure	17 (50.0)	–	17 (100.0)	–
Atrial septal defect closure	13 (38.2)	–	13 (76.5)	
Ventricular septal defect closure	3 (8.8)	–	3 (17.6)	
Aortic valve repair	1 (2.9)	–	1 (5.9)	
PAPVR repair	1 (2.9)	–	1 (5.9)	
RV infundibular muscle resection	1 (2.9)		1 (5.9)	
Cardiopulmonary bypass time (min)	125 [112–141]	124 [97–142]	126 [117–136]	0.717
Aortic cross-clamp time (min)	69 [62–83]	69 [59–75]	69 [66–87]	0.262

Continuous variables are presented as the median [interquartile range]; categorical variables are presented as n (%). AAOCA, anomalous aortic origins of a coronary artery; PAPVR, partial anomalous pulmonary venous return; RV, right ventricle.

Vienna, Austria), and Python version 3.9.7 (Python Software Foundation, Wilmington, DE, USA). Continuous variables are presented as the median with IQR, and categorical variables are presented as numbers and percentages. Categorical variables were analyzed using Fisher's exact test, and continuous variables were analyzed using the Mann-Whitney U test. All tests were 2-tailed, and $P<0.05$ was considered significant.

Results

Baseline Characteristics

Of the 34 patients in this study, 25 (73.5%) were male. Median patient age at the time of diagnosis and surgery was 12 years (IQR 5.3–15.4 years) and 13.4 years (IQR 7.4–16.4 years), respectively. The cohort included 2 cases of anomalous aortic origin of the left coronary artery (AAOLCA) and 32 cases of anomalous aortic origin of right coronary artery (AAORCA). Eleven (32.4%) patients had additional congenital heart disease, and 5 (14.7%) had a history of Kawasaki disease. Three (8.8%) patients were elite athletes. Among the 22 patients with symptoms, chest pain was the most common symptom (n=16), followed by chest discomfort (n=4). Prior to surgery, 6 patients had abnormal TMT findings. Of the 7 patients with abnormal findings on preoperative myocardial SPECT, 1 had AAOLCA and 6 had AAORCA. In the two patients with AAOLCA, a reversible perfusion decrease was observed in the apico-anterior wall. Among the patients with AAORCA, except for 1 patient whose abnormality was confirmed but not quantified by another institution, mild (n=3) and moderate (n=2) decreased perfusion was observed in the corresponding area of the right coronary artery (RCA). There were no significant differences in baseline characteristics between patients undergoing isolated vs. concomitant AAOCA repair, or between those undergoing unroofing vs. reimplantation (**Table 1**; **Supplementary Table 1**).

Operative Data

The most common surgical technique was unroofing (n=32; 94.1%). A full median sternotomy was performed in 30 (88.2%) patients. Resuspension sutures to prevent AR were placed in 9 (26.5%) patients. Concomitant procedures were performed in 17 (50.0%) patients, with closure of the atrial septal defect being the most frequent (n=13). The median cardiopulmonary bypass time was 125 min (IQR 112–141 min) and the median aortic cross-clamp time was 69 min (IQR 62–83 min). There were no significant differences in operative data between patients undergoing isolated vs. concomitant AAOCA repair, or between those undergoing unroofing vs. reimplantation (**Table 2**; **Supplementary Table 1**).

Early and Mid-Term Clinical Outcomes

The median postoperative hospital stay was 5 days (IQR 4–7 days). The median ICU stay was 23.4 h (IQR 21.5–25.9 h), and the median time to extubation was 4.5 h (IQR 3.4–6.3 h), except for 2 patients who were extubated in the operating room. No operative mortality or coronary artery-related complications occurred. Other complications were observed in 4 patients: 2 surgical wound infections, 1 respiratory complication, and 1 atrial flutter. Newly developed AR (trivial in 3 patients and mild in 1; 11.8%) was observed postoperatively, with 1 patient undergoing a resuspension suture procedure.

The median follow-up duration was 3.4 years (IQR 2.4–4.3 years; maximum 15.8 years). No deaths or coronary reinterventions were recorded during follow-up. Twelve (35.3%) patients reported symptoms at least once after surgery, primarily chest pain (which was presumed to be would pain) in 9 patients. However, 2 patients continued to experience symptoms by the last follow-up at 2 years (**Figure 2**). Postoperative cardiac CT revealed mild left main coronary artery stenosis in 1 patient with AAOLCA and mild RCA stenosis in 1 patient with AAORCA. Postoperative TMT (n=24), myocardial SPECT (n=9),

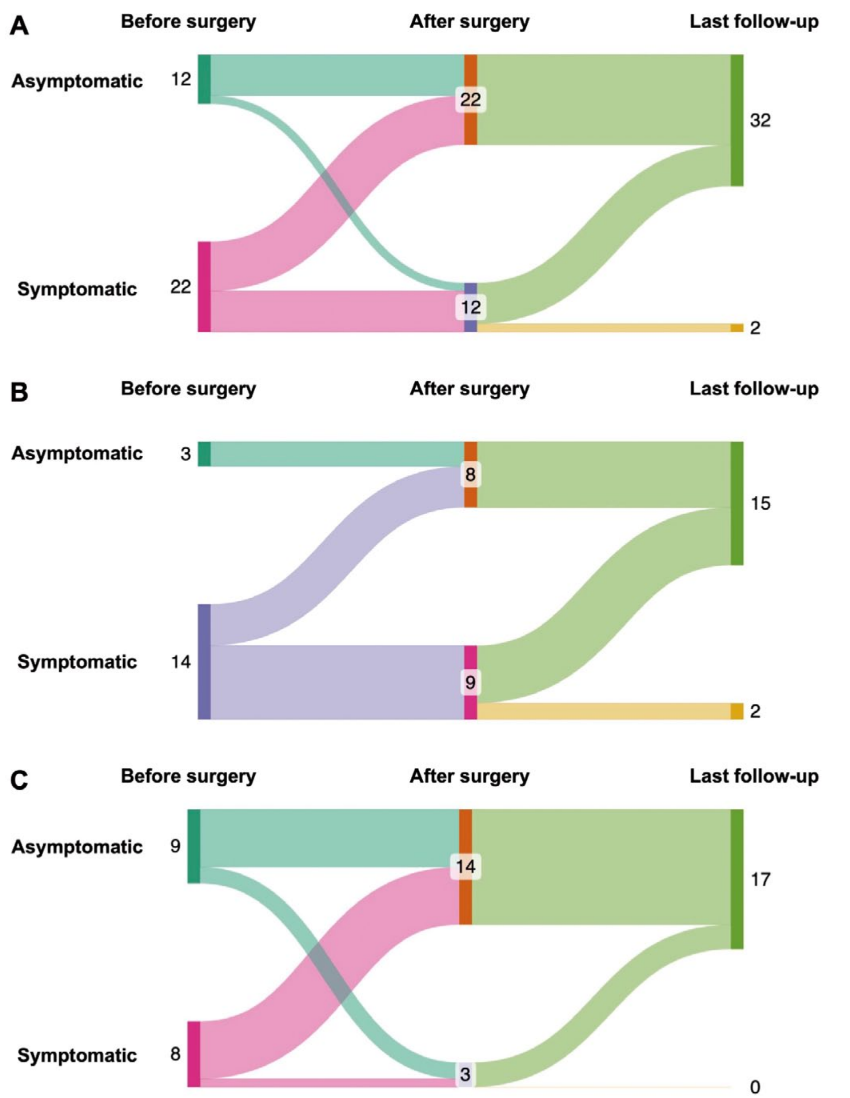


Figure 2. Transitions in patient symptoms from before surgery to after surgery and the last follow-up visit in (**A**) the overall cohort, (**B**) patients underdoing isolated anomalous aortic origin of the coronary artery (AAOCA) repair, and (**C**) patients underdoing concomitant AAOCA repair.

coronary angiography (n=2), and CMR (n=1) revealed no abnormalities, except in one patient who showed a reversible mild decrease in stress perfusion at the mid-basal inferoseptal wall on SPECT, improved from the moderate decrease observed preoperatively. There were no significant differences in early and mid-term clinical outcomes between those undergoing isolated vs. concomitant AAOCA repair, or between those undergoing unroofing vs. reimplantation (**Table 3**; **Supplementary Table 1**).

Perioperative Changes in Functional Measures and Symptoms

Of the 22 patients with symptomatic AAOCA, 8 had abnormal findings on preoperative functional studies, with the abnormal findings persisting in 1 patient postoperatively without symptoms. Of the 12 patients with asymptomatic AAOCA, 5 had abnormal findings on preoperative functional studies, with no abnormal findings detected postoperatively. However, one patient in the preoperative asymptomatic group demonstrated mild RCA stenosis postoperatively, which had not been detected in the preoperative studies (**Figure 3**).

Three patients had abnormal findings on postoperative examinations: 2 on imaging studies and 1 on functional studies. Two patients had preoperative chest pain, and 1 was asymptomatic. One symptomatic patient (Patient no. 27), a 17-year-old male who underwent an unroofing procedure for AAORCA, showed a moderate stress perfusion decrease in the entire inferior wall on preoperative myocardial SPECT, and his symptoms resolved after surgery. Postoperative SPECT revealed a mild, reversible stress perfusion decrease in the mid-basal inferoseptal wall, with residual stress perfusion improvement. Another symptomatic patient (Patient no. 2), a 4-year-old boy who underwent an unroofing procedure for AAOLCA without a preoperative functional study, had mild residual stenosis at the left main coronary ostium on cardiac CT 9.5 years postoperatively but remained asymptomatic with normal results on the TMT. Patient no. 3, an asymptomatic 4-year-old girl who underwent an unroofing procedure for AAORCA, also without a preoperative functional study, showed mild proximal RCA stenosis on postoperative CT.

Table 3. Early and Mid-Term Clinical Outcomes

	Total (n=34)	AAOCA repair		P value
		Isolated (n=17)	Concomitant (n=17)	
Early clinical outcomes				
Operative mortality	0 (0.0)	0 (0.0)	0 (0.0)	
Hospital stay (days)				
Total	7 [5–9]	8 [6–9]	6 [5–9]	0.375
After surgery	5 [4–7]	6 [5–7]	5 [4–6]	0.380
Intensive care unit stay (h)				
Total	23.4 [21.5–25.9]	23.7 [21.2–26.1]	23.2 [21.8–24.3]	0.654
Mechanical ventilation	4.5 [3.4–6.3]	4.6 [3.5–7.8]	4.4 [3.2–5.2]	0.535
Newly diagnosed aortic regurgitation	4 (11.8)	2 (11.8)	2 (11.8)	>0.999
Complication	4 (11.8)	1 (5.9)	3 (17.7)	0.601
Coronary artery-related complication	0 (0.0)	0 (0.0)	0 (0.0)	
Surgical wound infection	2 (5.9)	1 (5.9)	1 (5.9)	
Respiratory complication	1 (2.9)	0 (0.0)	1 (5.9)	
Atrial flutter	1 (2.9)	0 (0.0)	1 (5.9)	
Mid-term clinical outcome				
Follow-up duration (years)				0.007
Median [interquartile range]	3.4 [2.4–4.3]	4.2 [3.6–6.2]	3.1 [2.2–3.3]	
Range	1.6–15.8	2.0–15.1	1.6–15.8	
Death	0 (0.0)	0 (0.0)	0 (0.0)	
Reintervention for coronary artery	0 (0.0)	0 (0.0)	0 (0.0)	
Symptoms				
At least once after surgery	12 (35.3)	9 (52.9)	3 (17.7)	0.071
Chest pain	9 (26.5)	7 (41.2)	2 (11.8)	
Chest discomfort	1 (2.9)	1 (5.9)	0 (0.0)	
Dyspnea on exercise	2 (5.9)	1 (5.9)	1 (5.9)	
Persistent until the last follow-up	2 (5.9)	2 (11.8)	0 (0.0)	0.485
Cardiac CT	34 (100.0)	17 (100.0)	17 (100.0)	–
Abnormal	2 (5.9)	1 (5.9)	1 (5.9)	>0.999
CMR	1 (2.9)	0 (0.0)	1 (5.9)	>0.999
Abnormal	0 (0.0)	0 (0.0)	0 (0.0)	–
CAG	2 (5.9)	2 (11.8)	0 (0.0)	0.485
Abnormal	0 (0.0)	0 (0.0)	0 (0.0)	–
TMT	24 (70.6)	13 (76.5)	11 (64.7)	0.708
Abnormal	0 (0.0)	0 (0.0)	0 (0.0)	–
Myocardial SPECT	10 (29.4)	8 (47.1)	2 (11.8)	0.057
Abnormal	1 (2.9)	1 (5.9)	0 (0.0)	>0.999

Continuous variables are presented as the median [interquartile range]; categorical variables are presented as n (%). AAOCA, anomalous aortic origins of a coronary artery; CAG, coronary angiography; CMR, cardiac magnetic resonance imaging; CT, computed tomography; SPECT, single-photon emission computed tomography; TMT, treadmill test.

Despite initial chest pain after surgery, her symptoms resolved, with no significant findings on TMT or myocardial SPECT (**Table 4**).

Of the patients with abnormal findings on preoperative TMT or myocardial SPECT, 84.6% (11/13) demonstrated improvement on postoperative functional evaluation. In detail, all five patients with abnormal preoperative TMT findings and 6 of 8 (75.0%) with abnormal preoperative myocardial SPET findings showed improvement (**Figure 4**).

Discussion

The prevalence of AAOCA ranges from 0.1% to 0.3% of the population, based on previous reports from the Europe, with AAORCA being 6- to 10-fold more common than AAOLCA.[11] Quantifying the increased risk of sudden cardiac death from AAOCA is challenging. However, studies involving large cohorts, including military recruits and young athletes, indicate that AAOLCA contributes to 2–5% of cases of sudden cardiac death, whereas AAORCA accounts for only 0.1–0.2% of cases.[12] Histological evidence from autopsies indicates that chronic ischemia due to an inadequate blood supply from the anomalous coronary artery leads to myocardial necrosis and fibrosis, potentially resulting in fatal ventricular arrhythmias.[13]

A previous multicenter study of 560 patients identified specific anatomic features that increase the risk of sudden cardiac death, such as an intramural course and a high or slit-like coronary orifice for AAOLCA, and a longer intramural course for AAORCA.[14] Recent guidelines similarly

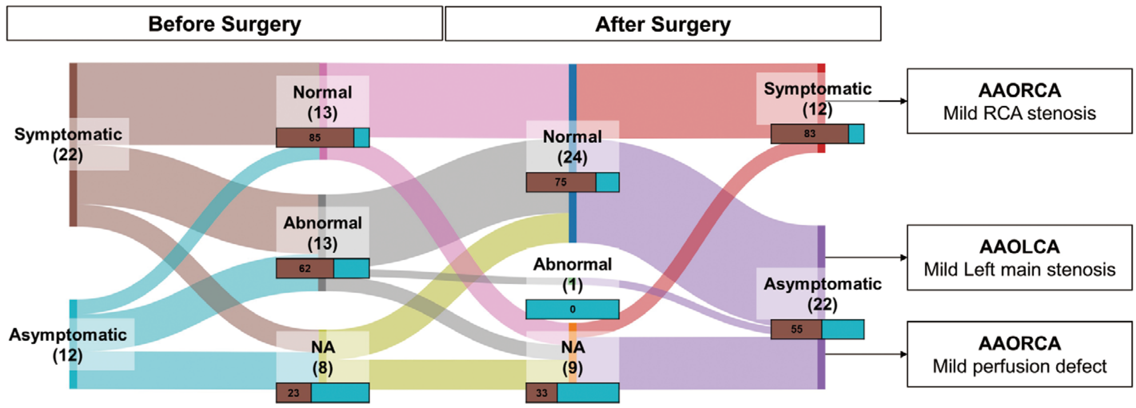


Figure 3. Perioperative changes in functional investigations and postoperative symptoms. Functional investigations were classified as "abnormal" if the treadmill test (TMT) or myocardial single photon emission computed tomography (SPECT) showed abnormalities. Patients reporting symptoms at any point after surgery were categorized as symptomatic. The number of subjects is denoted in parentheses. In each node, the box below illustrates the composition of patients based on preoperative symptoms. The red segment indicates the proportion of patients who were symptomatic before surgery, and the blue segment indicates those who were asymptomatic. The number inside each box represents the percentage of patients with preoperative symptoms within that node. AAOLCA, anomalous aortic origin of the left coronary artery; AAORCA, anomalous aortic origin of the right coronary artery; NA, not available; RCA, right coronary artery.

Table 4. Details of Patients With Abnormal Findings on Postoperative Investigations

Patient no.	Type of AAOCA	Postoperative symptoms	Postoperative investigations			Follow-up duration (years)
			Method	Time after surgery	Finding	
2	AAOLCA	None	CT	9.5 years	Suspicious mild residual stenosis of the left main coronary ostium	15.1
			TMT	14.9 years	Normal	
3	AAORCA	Chest pain	CT	13.6 years	Mild stenosis at the proximal right coronary artery	15.8
			TMT	13.4 years	Normal	
			Myocardial SPECT	13.6 years	Normal	
27	AAORCA	None	CT	6 days	Patent right coronary artery without stenosis	2.2
			TMT	0.5 year	Normal	
			Myocardial SPECT	0.6 year	Mild stress perfusion decrease, reversible at rest	

AAOCA, anomalous aortic origin of a coronary artery; AAOLCA, anomalous aortic origin of the left coronary artery; AAORCA, anomalous aortic origin of the right coronary artery; CT, computed tomography; SPECT, single photon emission computed tomography; TMT, treadmill test.

highlight high-risk anatomies that could compromise coronary perfusion, including an intramural course, fish mouth-shaped or slit-like orifices, acute-angle takeoffs, or high take-offs.[4,5] In the present study, all patients had an interarterial course (between the aorta and main pulmonary artery) and, except for 1 patient with only a high takeoff, all had an intramural course (**Supplementary Table 2**).

Coronary reimplantation or unroofing for AAOCA rarely results in operative mortality and is associated with symptom improvement and satisfactory short-term clinical outcomes based on postoperative echocardiographic data.[15–18] Our findings align with the existing literature, with no deaths recorded and no need for coronary reintervention during the follow-up period. However, a previous study indicated an 8% incidence of new mild or severe AR, a non-negligible surgical complication.[19] To prevent AR, we do not unroof the entire intramural course, preserving the aortic valve commissure and creating double openings in patients in whom the intramural course is located below the commissural level.[9] In our study, 1 patient with congenital aortic stenoinsufficiency underwent aortic valve repair with leaflet slicing and valvotomy, reducing AR from mild to trivial. We observed only 1 case of newly developed mild AR, which is encouraging compared with previous studies.

Recent guidelines highlight the importance of preoperative physiological or functional tests, such as nuclear studies, to confirm or exclude stress-induced ischemia, but they do not specify follow-up methods after surgery.[4,5,20] The 2017 American Association for Thoracic Surgery expert guidelines recommend a normal exercise stress test before resuming competitive sports, emphasizing the importance of postoperative functional tests.[21] Moreover, several studies demonstrating perioperative changes in

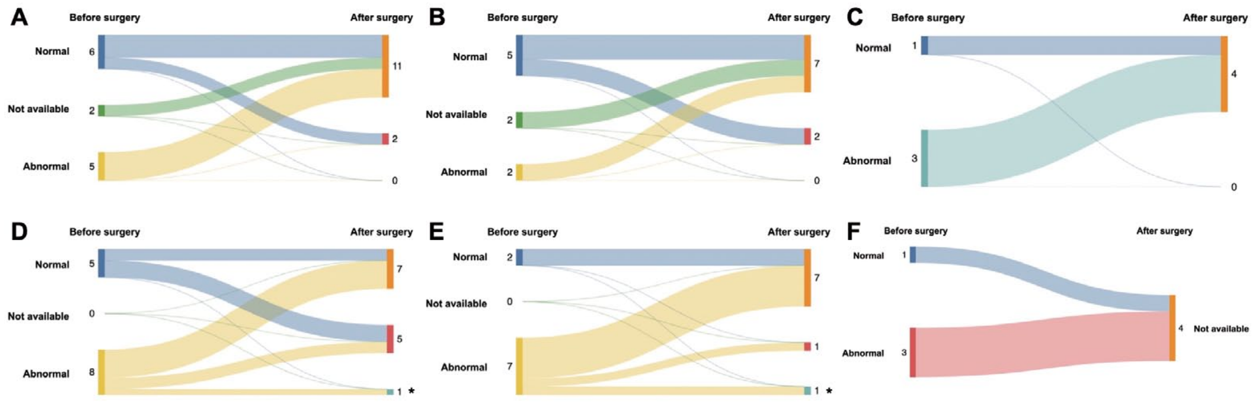


Figure 4. Perioperative changes in functional investigations: (**A**) treadmill test (TMT), overall cohort; (**B**) TMT, isolated anomalous aortic origin of a coronary artery (AAOCA) repair; (**C**) TMT, concomitant AAOCA repair; (**D**) myocardial single photon emission computed tomography (SPECT), overall cohort; (**E**) myocardial SPECT, isolated AAOCA repair; and (**F**) myocardial SPECT, concomitant AAOCA repair. *Although the postoperative examination for this patient is categorized as abnormal, it actually represents an improvement. Preoperatively, there was a moderate decrease in stress perfusion involving the entire inferior wall, but postoperatively the finding improved to a mild decrease in stress perfusion, which was reversible at rest and localized to the mid-basal inferoseptal wall.

functional investigations, such as exercise stress tests and myocardial perfusion, underscore the need for serial follow-up with postoperative functional studies.[21–23] In our study, all patients with preoperative abnormalities on TMT or myocardial SPECT showed improvement after surgery, except for 1 patient who did not undergo a postoperative functional study (**Table 4**).

Postoperative symptoms, such as chest pain, may persist but should not be hastily attributed to compromised coronary perfusion.[18,23] In our study, 12 patients reported symptoms after surgery, but only 2 continued to experience symptoms until the last follow-up, with none exhibiting abnormal findings on functional studies (**Figure 2**). This suggests most postoperative chest pain is likely due to surgical wound pain rather than coronary problems. Two patients had findings suggestive of ostial stenosis on postoperative CT, although it remains unclear whether this mild stenosis is related to the surgical technique. Both patients were discharged after confirming patent coronary flow and the absence of symptoms on echocardiography prior to discharge. For the patient with AAOLCA, although asymptomatic during follow-up, CT performed 9.6 years after surgery as part of routine follow-up revealed a suspicious finding in the left main coronary artery. However, subsequent echocardiography 15.3 years after surgery showed patent coronary flow, with normal findings on a TMT 15.1 years after surgery. For the patient with AAORCA, follow-up echocardiography at 5.9 years showed patent coronary flow without symptoms. However, 13.6 years after surgery, the patient reported chest pain and thorough evaluations were conducted. CT suggested suspicious stenosis near the ostium of the AAORCA, but neither SPECT nor TMT revealed any abnormalities. At 15.1 years, echocardiography confirmed patent coronary flow. The chest pain subsided following the evaluations and was ultimately considered unlikely to be of coronary etiology; rather, it was more likely attributable to postoperative wound pain (**Table 4**). Both patients have remained asymptomatic without evidence of disease progression to date. The perioperative changes in functional studies and the resolution of symptoms suggest favorable mid-term clinical outcomes and underscore the potential utility of functional examinations as a follow-up tool, although further long-term evaluation is warranted.

Enhancing the diagnostic rate is as important as the introduction of safe and effective surgical techniques. In our study, the median age at diagnosis of AAOCA was 12.0 years, suggesting that in pediatric patients AAOCA is often only suspected and investigated when symptoms arise. Notably, 12 (35.3%) asymptomatic patients in our cohort were diagnosed incidentally with AAOCA during evaluations for other cardiac conditions, such as ventricular septal defect, atrial septal defect, or Kawasaki disease. Therefore, in asymptomatic patients, it is essential to establish a standardized echocardiographic protocol to ensure that AAOCA is not overlooked during evaluations for other cardiac diseases.[24] Furthermore, in patients presenting with occasional atypical chest pain, it may be necessary to consider CT angiography to rule out AAOCA. Effectively identifying what can already be observed is crucial, but for patients who are asymptomatic or who do not have any concurrent cardiac conditions, future studies could explore the development of risk models using artificial intelligence that integrate data from electrocardiograms obtained using mobile devices, family history, and other clinical records.[25]

We suggest that surgical treatment should be considered even in patients with asymptomatic AAORCA if they exhibit high-risk anatomy for 3 reasons. First, in our study, 12 patients with asymptomatic AAORCA were identified, of whom 5 (41.7%) had abnormal preoperative functional examination findings. This underscores the need for a thorough evaluation of incidentally detected AAORCA, because current guidelines recommend considering surgery if functional studies reveal abnormalities.[4,5] Second, 41.7% (5/12) of patients with asymptomatic AAORCA had a history of Kawasaki disease. Of such patients, approximately 30–50% experience transient coronary artery enlargement during the acute phase, and, if untreated, up to 25% may progress to severe coronary abnormalities, such as aneurysms or ectasia.[26–28] These abnormalities significantly increase the risk of coronary thrombosis, stenosis, myocardial infarction, and sudden death years after the diagnosis, making AAORCA a more problematic condition.[28,29] The risk of sudden cardiac death may be further increased in patients with asymptomatic AAORCA and a history of Kawasaki disease. Our data suggest that, in Asia, where the prevalence of Kawasaki disease is relatively high, a different approach for managing asymptomatic AAORCA may be warranted.[30] Third, although the risk of sudden cardiac death in AAORCA is low, it is not negligible. Our study included 3 elite athletes with symptomatic AAORCA who resumed unrestricted sports after surgery. Most of the other patients were students who enjoyed various competitive sports after surgery without concerns regarding coronary problems.

For these reasons, Seoul National University Children's Hospital adopts a proactive approach to surgical intervention, even in cases of incidentally detected AAORCA, particularly when high-risk anatomical features are present. This approach may explain, in part, the predominance of AAORCA in our cohort. In particular, for active individuals who engage in vigorous physical activities such as swimming, football, or basketball, we tend to recommend surgical correction given its potential to enhance both safety and quality of life. After successful surgery, we are able to encourage these patients to continue participating in their preferred activities without concern for coronary complications. This is especially important for younger patients (children and adolescents) because unrestricted participation in daily life and sports plays a vital role in their physical, emotional, and psychosocial development. If surgery can be performed safely with minimal risk, it should be actively discussed, taking into account the patient's circumstances and quality of life.[11]

Study Limitations

This study has 2 limitations. First, it was a retrospective single-center study with a small sample size. In our study, AAORCA was observed in 94.1% of patients. The proportion of surgical intervention observed in this study is higher than would generally be expected, particularly considering that sudden cardiac death has been more frequently reported in patients with AAOLCA, who more often require surgical management. This discrepancy is likely attributable to the small sample size and institutional referral patterns in our cohort. Therefore, caution is warranted in interpreting our findings; expanding the sample size or conducting multicenter studies may help mitigate this selection bias.[11,12] Second, this study evaluated only early and mid-term clinical outcomes. Findings from previous studies suggest that subclinical ischemia and chronotropic impairment may persist or emerge even after successful AAOCA repair, potentially affecting long-term outcomes.[22,31] Therefore, further extended follow-up is essential to comprehensively assess the long-term impact of surgical repair, particularly identifying latent ischemic changes and potential declines in exercise capacity, especially considering that the majority of patients were still in the growth phase at the time of this study.

Conclusions

Surgical repair of AAOCA is safe and effective, with acceptable early and mid-term outcomes. Surgery should be considered for patients with asymptomatic AAORCA and high-risk coronary anatomical features.

Acknowledgments

During the preparation of this manuscript, the authors used ChatGPT 4.0 in order to correct grammatical errors. After using ChatGPT 4.0, the authors reviewed and edited the content as required and take full responsibility for the content of the publication.

Sources of Funding

This study was supported by a Seoul National University Research Grant (No. 800-20230598).

Disclosures

None.

IRB Information

The study protocol was reviewed by the Institutional Review Board of Seoul National University Children's Hospital was approved as a minimal-risk retrospective study (H-2308-094-1458, August 21, 2023). The requirement for individual consent was waived.

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Supplementary Files

Please find supplementary file(s);
https://doi.org/10.1253/circj.CJ-25-0097